

MECHANISM OF MUSCLE CONTRACTION

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 - ENERGY FOR MUSCULAR CONTRACTION
 - ▼ RELAXATION OF MUSCLE
 - Sequence of Events During Muscular Relaxation:

- **1. Physical Change :—** Change in length of muscle fibers / change in tension developed in muscle.
- **2. Chemical Change :—** Energy necessary for muscular contraction is liberated during processes of breakdown & resynthesis of ATP.
- **3. Histological Change :—**
 - a. **Molecular Basis of Muscular Contraction :** Molecular mechanism responsible for formation of actomyosin complex (includes 3 stages: Excitation–Contraction Coupling, Role of Troponin & Tropomyosin, Sliding Mechanisms).
 - b. **Actomyosin Complex :**
 - In relaxed state of muscle → Thin actin filaments from opposite ends of sarcomere are away from each other, leaving a broad 'H' zone.
 - During contraction of muscle → Actin filaments glide over myosin to form **actomyosin complex**.

1. EXCITATION–CONTRACTION COUPLING

- **Definition :** Process that occurs between excitation and contraction of muscle.
- Involves a series of activities responsible for contraction of excited muscles.

Stages of Excitation–Contraction Coupling:

- Ca^{2+} ions from sarcoplasm move towards actin filaments to produce contraction.
- Ca^{2+} **ions form the link or coupling material** between excitation and contraction of muscle.

Excitation of muscle fibers
by impulses passing
through motor nerve and
neuromuscular junction



Generation of action
potential in muscle fiber



Spread of action potential
via 'T' tubules



Entrance of action
potential in cisternae



Release of calcium ions
from cisternae



Movement of calcium ions
towards actin filaments



Contraction of the muscle
fibers

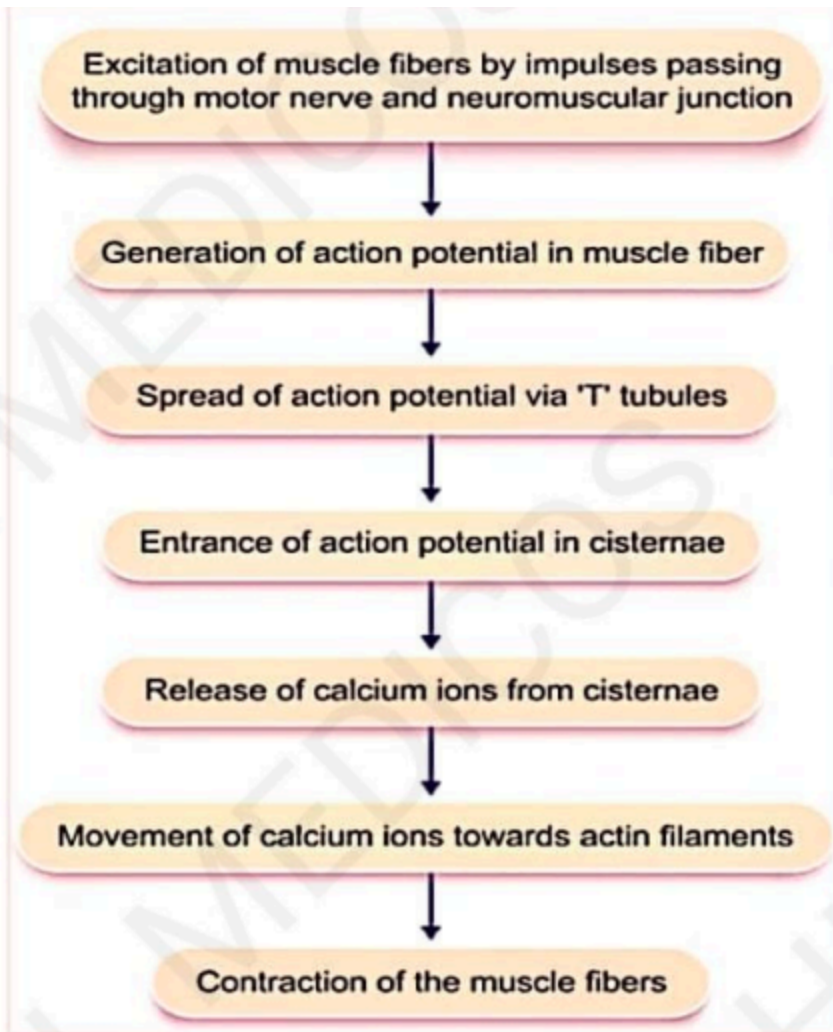


FIGURE 31.6: Excitation-contraction coupling

2. ROLE OF TROPONIN AND TROPOMYOSIN

- **Head of myosin molecules** has a strong tendency to get attached with active site of F-actin.
- **In relaxed condition** : Active site of F-actin is covered by **tropomyosin**.

Ca²⁺ ions released from 'L'
tubules during excitation
of muscle



Bind with Troponin C



Loading of Troponin C with
Ca²⁺ ions produces change
in position of troponin
molecules



Pulls tropomyosin
molecule away from F-
actin



Active site of F-actin
uncovered & exposed



Immediately, head of
myosin gets attached to
actin

3. SLIDING MECHANISM

- **Formation of Actomyosin Complex** → **Sliding Theory** :

Explains how actin filaments slide over myosin filaments and form actomyosin complex during muscular contraction (*also called **Ratchet Theory** / **Walk-Along Theory***).

- **Each cross-bridge from myosin filaments has 3 components** :—

- a. Hinge
- b. An arm
- c. A head

- **Power Stroke** : Myosin head is tilted towards arm so that actin filament is dragged along with it. Tilting of head is called **power stroke**.

Detailed Steps of Sliding Mechanism:

Release of large amount of
calcium ions from
cisternae of 'L' tubules



Binding of calcium ions
with troponin



Change in position of
troponin



Pulling of tropomyosin
molecule away from F actin



Exposure of active sites of
F actin



Attachment of myosin
head to actin



Power stroke - tilting of
myosin head and dragging
of actin filament



Detachment of myosin

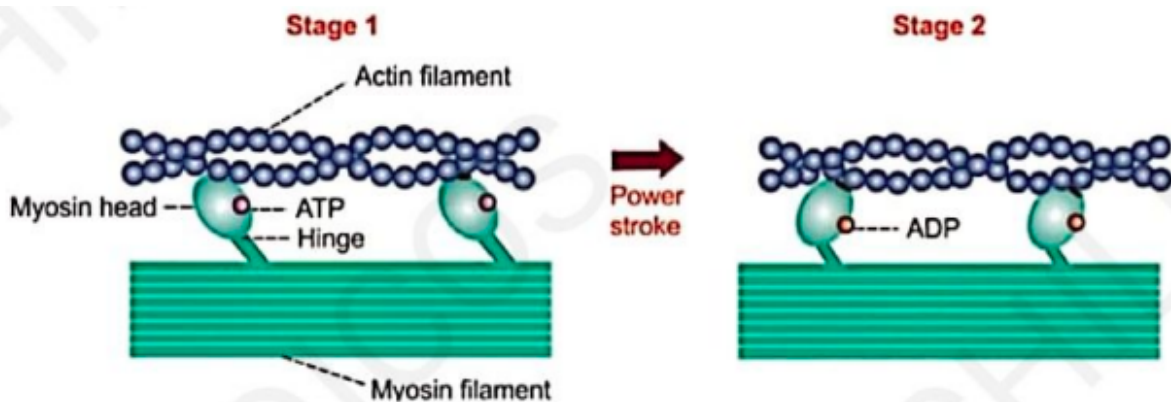
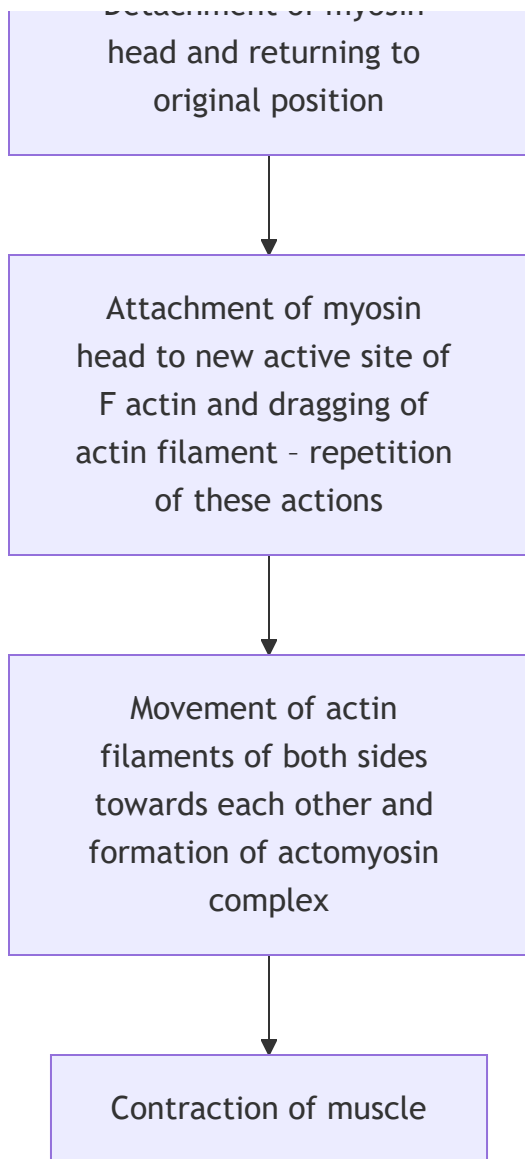


FIGURE 31.7: Diagram showing power stroke by myosin head.
Stage 1: Myosin head binds with actin; Stage 2: Tilting of myosin head (power stroke) drags the actin filament.

CHANGES IN SARCOMERE DURING MUSCULAR CONTRACTION

- a. **Length of all sarcomeres decreases** as 'Z' lines come closer.
- b. **Length of all 'I' bands decreases** since actin filaments from opposite sides overlap.
- c. **'H' zone either decreases or disappears.**
- d. **Length of 'A' band remains same.**

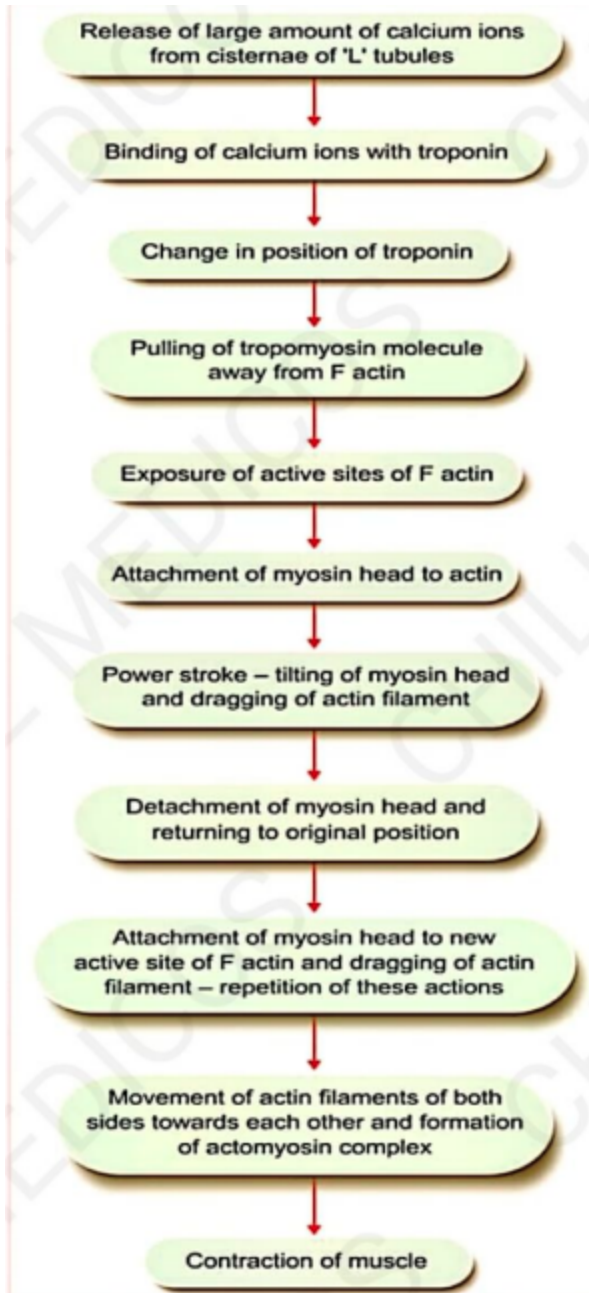
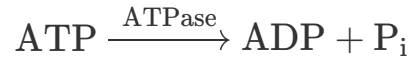


FIGURE 31.8: Sliding mechanism

ENERGY FOR MUSCULAR CONTRACTION

- **Energy for movement of myosin head (power stroke) :—**
Obtained by breakdown of ATP into ADP + P_i.
- Head of myosin has a **site for ATP**; head itself can act as enzyme **ATPase**.
- **Sequence of Events :—**
 - a. When tropomyosin moves to expose active sites → Head attaches to active site:

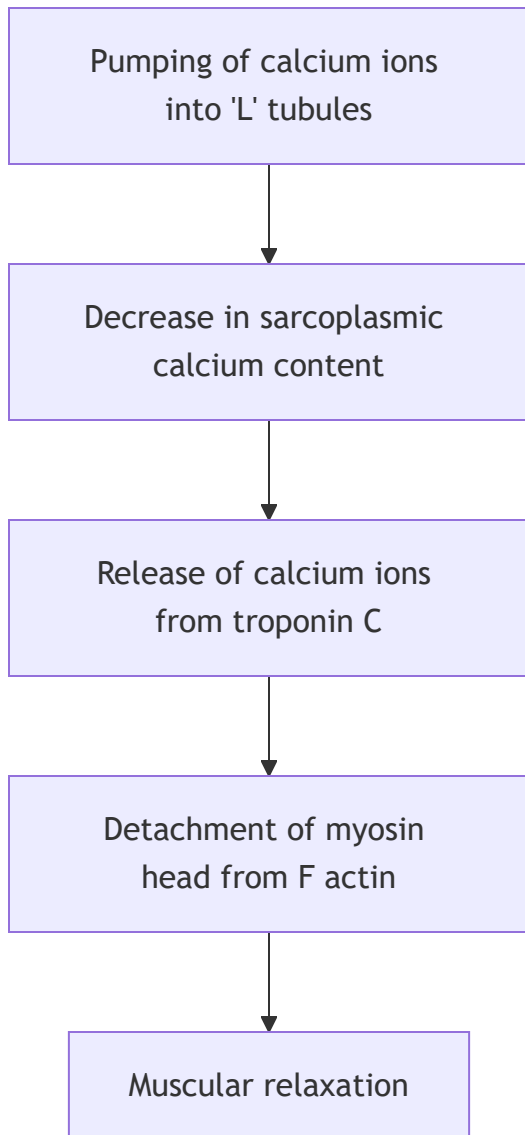


- b. Energy released is utilized for contraction.
- c. When head is tilted, ADP + P_i is released and **new ATP molecule binds with head**.

RELAXATION OF MUSCLE

- **Relaxation occurs** when Ca²⁺ ions are pumped back into L-tubules.
- **Detachment of myosin from actin** obtains energy from breakdown of ATP.
- **Muscular relaxation is an ACTIVE process.**

Sequence of Events During Muscular Relaxation:



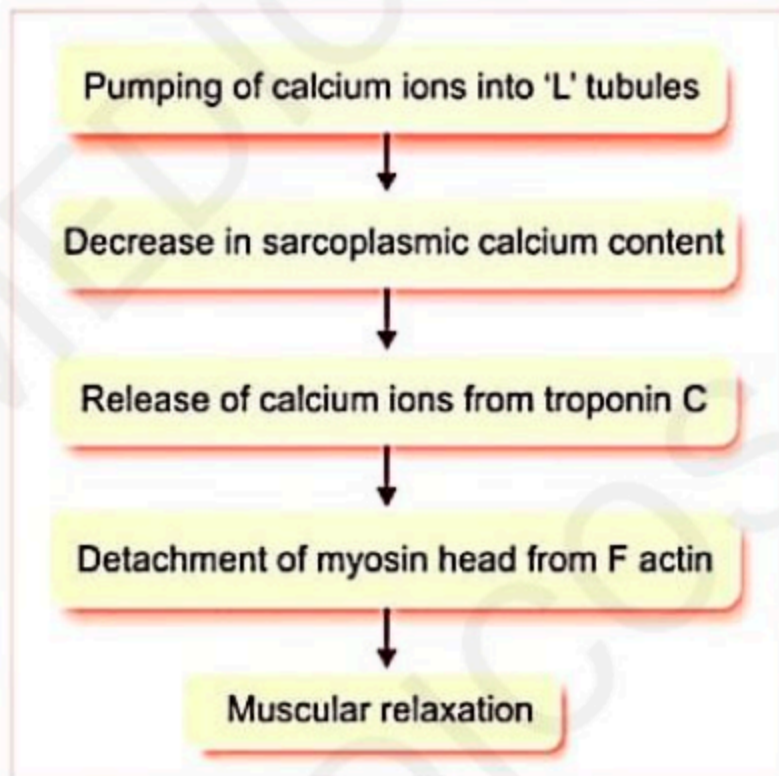


FIGURE 31.9: Sequence of events during muscular relaxation